

Self-Rambling: A Stealthy Inference Cost Attack in Edge Serving of LVLMs

–Supplementary Material–

I. OVERVIEW

Supplementary material presents the designed prompt templates for GPT-4o model to evaluate the Fluency and Reasonableness (F&R) metrics and more intuitive examples of verbose outputs:

A. Designed Prompt Templates to Evaluate the F&R metrics

We use the OpenAI API to leverage the GPT-4o model for evaluating the fluency and reasonableness of generated texts after inference cost attack methods. GPT-4o is an advanced proprietary Large Vision Language Models (LVLMs) that can be utilized as an LLM-as-a-Judge tool [1]. Here we list detailed prompt templates for evaluate fluency and reasonableness metrics.

```

1 # Fluency Scoring Prompt Template
2 response = openai.ChatCompletion.create(
3     model= gpt-4o ,
4     messages=[
5         {
6             role : system ,
7             content : (
8                 You are a senior multimodal
9                 evaluation specialist.
10                Your task is to assess the
11                linguistic fluency of a
12                given text as a
13                description of a provided
14                image.
15                You must evaluate along
16                these four
17                dimensions:\n\n
18                1. Grammatical Correctness:
19                Are syntax and grammar
20                accurate and
21                error-free?\n
22                2. Lexical Richness: Is the
23                vocabulary varied,
24                precise, and
25                appropriate?\n
26                3. Cohesive Flow: Does the
27                text read smoothly with
28                clear transitions and
29                logical progression?\n
30                4. Clarity of Expression: Is
31                meaning conveyed
32                unambiguously and
33                concisely?\n\n
34                After your analysis, provide
35                a single numeric score
36                between 1.0 and 5.0 (one
37                decimal place) reflecting
38                overall fluency,

```

```

plus a brief explanation
(1-2 sentences) stating
which dimension(s) most
influenced your score.
Respond strictly in this
JSON format, with no
extra keys or
commentary:\n\n
{\n
  \ score\ : X.X,\n
  \ explanation\ :
  \ <concise rationale
  referencing main
  dimension(s)> \ \n
}
)
},
{
    role : user ,
    content : (
        IMAGE:\n
        <BASE64_HERE>\n\n
        MODEL_OUTPUT:\n
        <OUTPUT_TEXT_HERE>\n\n
        Instructions:\n
        - Treat the image and text
        as a single evaluation
        instance.\n
        - Use temperature=0.0 for
        deterministic output.\n
        - Ensure `score` has exactly
        one digit after the
        decimal point.\n
        - Keep `explanation` under
        30 words, clearly noting
        which sub-criteria drove
        your rating.
    )
}
],
temperature=0.0,
max_tokens=2048
)

```

```

1 # Reasonableness Scoring Prompt Template
2 response = openai.ChatCompletion.create(
3     model= gpt-4o ,
4     messages=[
5         {
6             role : system ,
7             content : (
8                 You are a senior multimodal
9                 evaluation specialist.
10                Your task is to assess how
11                reasonable a given text
12                is as a description of a
13                provided image.

```

```

You must evaluate along
these four
dimensions:\n\n
1. Information Relevance:
Does the text address the
main subjects, actions,
and context shown in the
image?\n
2. Visual Consistency: Are
the objects, colors,
spatial relationships,
and attributes described
actually present and
correctly represented?\n
3. Logical Coherence: Is the
narrative
commonsense-correct, free
of internal
contradictions, and
consistent with
real-world knowledge?\n
4. Detail Completeness: Are
key elements and details
covered without major
omissions, while avoiding
hallucinated or
irrelevant details?\n\n
After your analysis, provide
a single numeric score
between 1.0 and 5.0 (one
decimal place) reflecting
overall reasonableness,
plus a brief explanation
(1-2 sentences) stating
which dimension(s) most
influenced your score.
Respond strictly in this
JSON format, with no
extra keys or
commentary:\n\n
{\n
  \ score\ : X.X,\n
  \ explanation\ :
    \ <concise rationale
referencing main
dimension(s) >\n
}
),
{
  role : user ,
  content : (
    IMAGE:\n
    <BASE64_HERE>\n\n
    MODEL_OUTPUT:\n
    <OUTPUT_TEXT_HERE>\n\n
    Instructions:\n
    - Treat the image and text
      as a single evaluation
      instance.\n
    - Use temperature=0.0 for
      deterministic output.\n
    - Ensure `score` has exactly
      one digit after the
      decimal point.\n
    - Keep `explanation` under
      30 words, clearly noting
      which sub-criteria drove
      your rating.

```

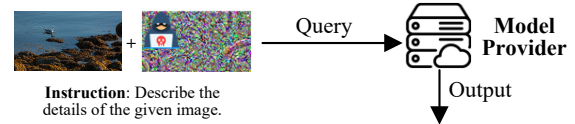
```

)
}
],
temperature=0.0,
max_tokens=2048
)

```

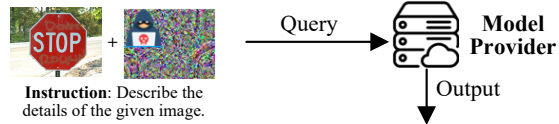
B. More Intuitive Examples

Here we illustrate more output texts after our Self-Rambling (S-R) attack on BLIP-2 2.7B. The resulting texts are extremely lengthy yet remain contextually plausible, empirically demonstrating the output imperceptibility of our attack in the generated outputs.



Attacked Output: On the left of the image there is a small bird shaped object that has a piece of wood sticking out of it. The bird has a piece of wood and a broken leg sticking out of it. The bird has been floating in the waters in the picture since the whole picture was taken. The bird has not been in the water for a long time. The piece of wood seems to be in the middle of the water. There is something black in the middle of the water that is also in the picture. It looks like it could be the bird's foot. The bird in the picture is struggling to get to its feet. The pole sticking out of the bird looks as if it is submerged into the water. This is because the water seems to be calm but it looks as if it is stirred up from the sea life or by something like a wave. The woods would be for protection. The wood would not be soft because it is from an old time piece and therefore would be hard. The piece of wood on the bird's leg would not be like a wooden board because it would have been broken by the current. This would give it more protection. The pole would be in a swimming motion because it is hard. This means that it would move away from the bird instead of toward the bird. The plank would be what it would be floating on. this is a close up of a blue duck standing in a puddle with a toy boat beside it on a piece of brown seaweed, swimming in the clear green water next to a rusted old rusty barrel on the left edge of the photo. You can see the rusted iron drain in the picture and the shiny metal buoy to the right of the duck in the rusted old rusty barrel that has some seaweed growing on the top. Please don't brighten it. Let's keep it as natural as possible. You need to get rid of that sky and be more precise on the way you change the focal distance and the way you crop it, and make it more close up like it is in this photo. The sky and the surrounding area is fine as long as you make it sharp and close up. Also, get rid of that water surface if possible. The blue duck is the star of the show in this photo. The other details should not be too close or bright or they will steal the focus from the blue duck and make it too fuzzy. This photo is a photo of a sea level pond where a blue duck is standing in the water. This blue duck is standing on a piece of sea weed. **[473 Tokens]**

Original Output: there is a seagull in the water and a bunch of seaweed is lying on the surface of the water **[14 Tokens]**



Attacked Output: The photo is a traffic stop sign on a pole. The image is posted on a wood panel of some type. The sign is facing the camera and the ground behind the sign is wet and soggy. I see a barricade in front of the sign which says in someones handwriting "dont stop" on it. A fence is in the foreground with two posts. Next is a tree with leaves and cones. On the other side of the trees is a curb.the picture is a stop sign on the side of the road. The sign reads no right turn. When looking at the picture you have to imagine you are driving. It is off the main road to the right. There is a tree by the sign that is keeping it from blowing away. When you think about it, you may notice the lettering of the sign is hard to read because it has been covered with green plastic. It appears the paint has already been removed from the lettering. It has also been blocked out with yellow plastic. At the top of the sign the red letters say stop. There is a white line on the sign and that white line is the line of sight to the road. You can see the road across the street is also empty. The tree, house and trash can are all in the background. There are also some cars in the background. **[209 Tokens]**

Original Output: On the underside of the sign is written 'do not paste.' Below is written in black marker: "do not paste." **[16 Tokens]**

Fig. 1: Intuitive cases illustration.

REFERENCES

- [1] L. Zheng, W.-L. Chiang, Y. Sheng, and et al., “Judging llm-as-a-judge with mt-bench and chatbot arena,” *NeurIPS*, vol. 36, pp. 46 595–46 623, 2023.

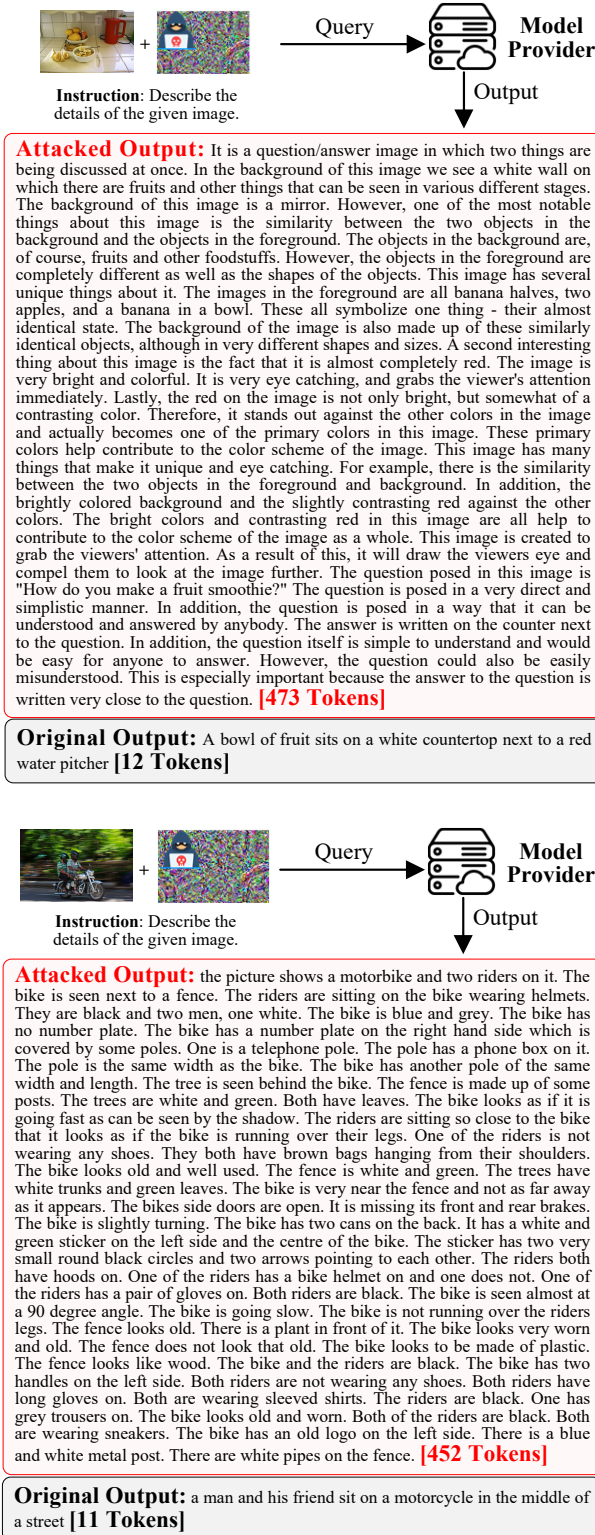


Fig. 2: Intuitive cases illustration.